AWS Auto Scaling – Step-by-Step Documentation

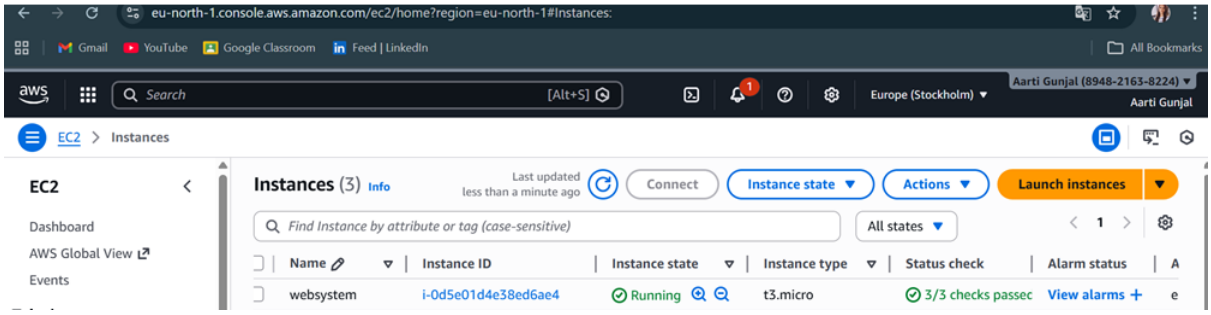
Auto Scaling is an AWS service that automatically adds or removes EC2 instances based on system load like CPU usage, traffic, or demand. It ensures that the application always has the right number of servers running.

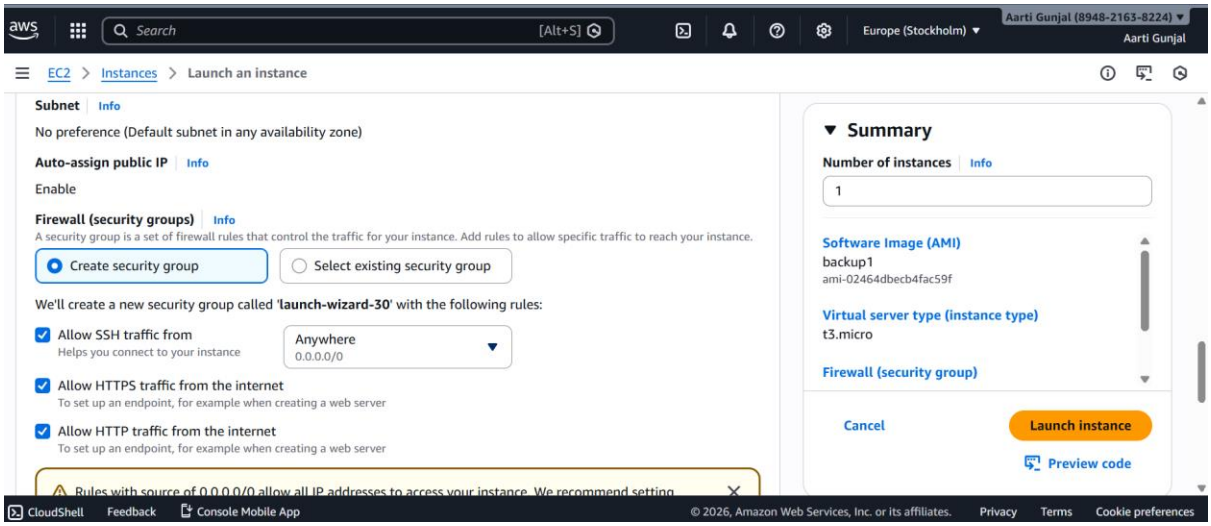
1. Create an EC2 Instance (Base Instance)

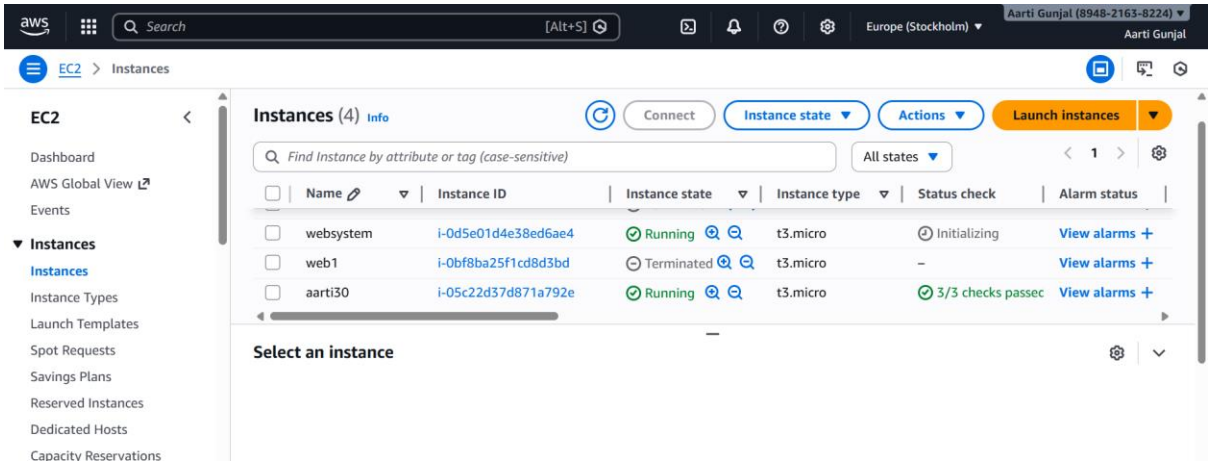
This instance will act as the **base server** for your application.

Steps:

1. Login to **AWS Management Console**
2. Go to **EC2 → Instances → Launch Instance**
3. Choose an **AMI** (Example: Amazon Linux 2)
4. Choose an **Instance Type** (Example: t2.micro – free tier)
5. Configure **Key Pair** (create or select existing)
6. Configure **Security Group**
   - Allow HTTP (Port 80)
   - Allow SSH (Port 22)
7. Launch the instance





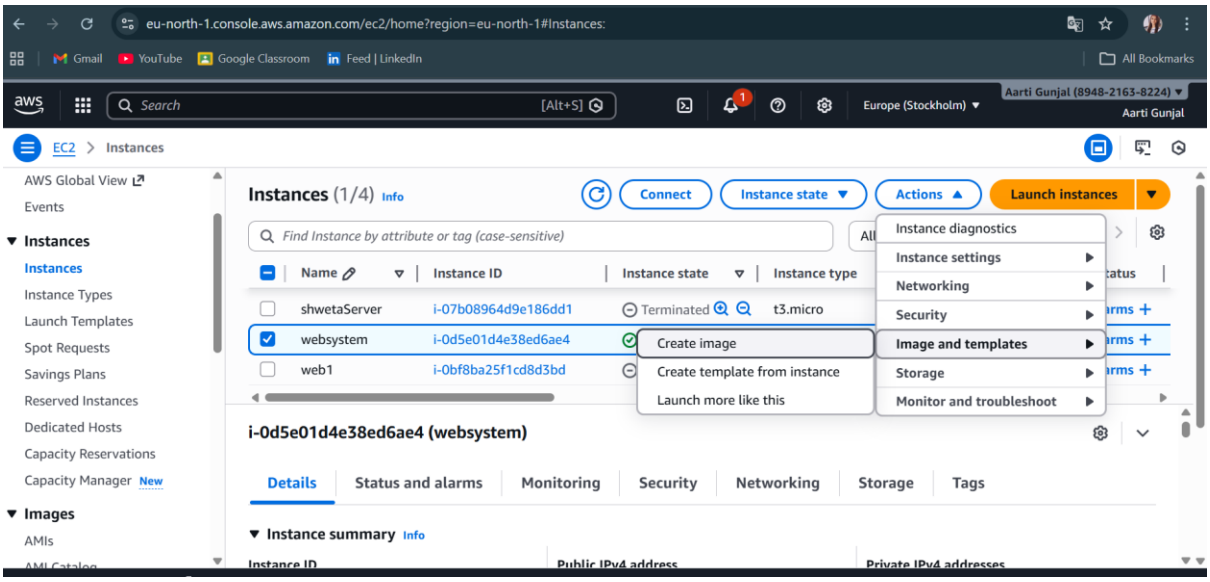


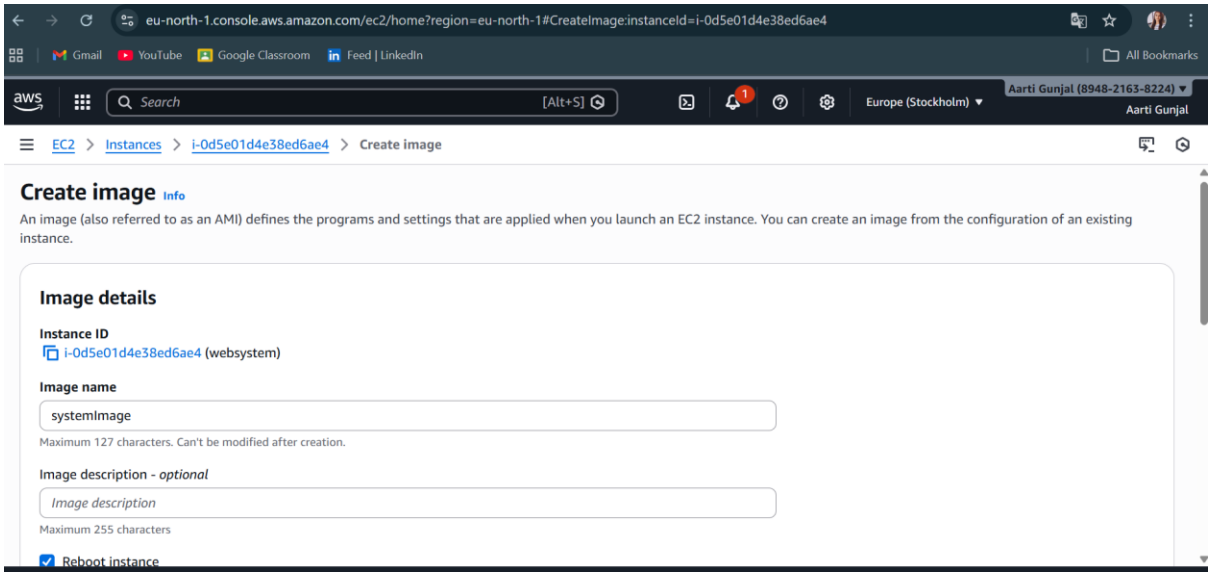
2. Create an AMI (Amazon Machine Image)

AMI is used to launch identical EC2 instances automatically.

Steps:

1. Go to **EC2 → Instances**
2. Select your configured EC2 instance
3. Click **Actions → Image and Templates → Create Image**
4. Enter:
   - Image Name
   - Image Description
5. Click **Create Image**




eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateImage:instanceId=i-0d5e01d4e38ed6ae4
Gmail
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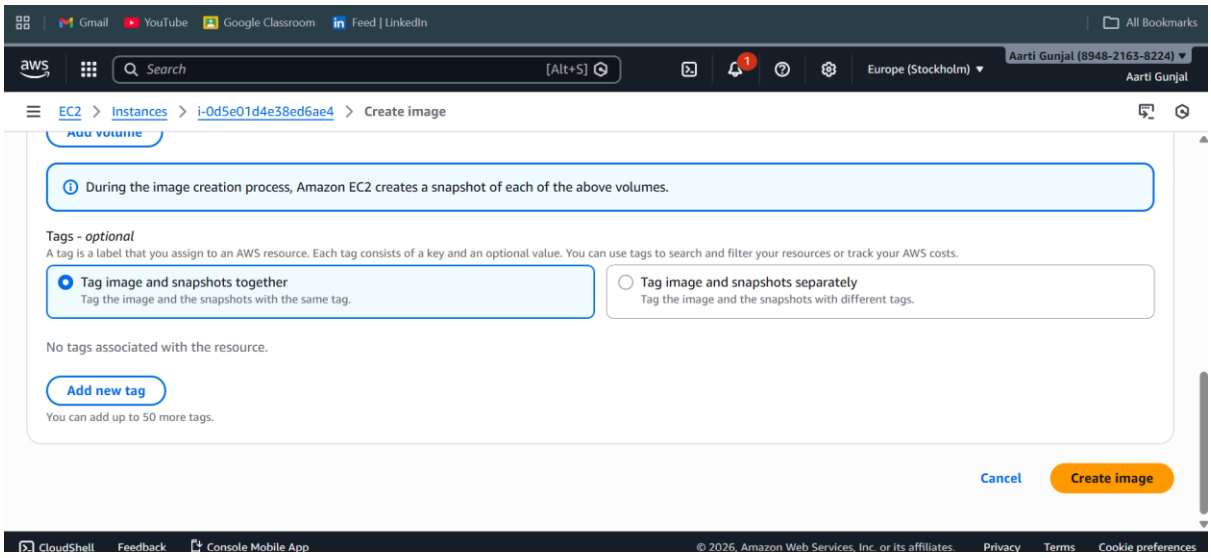
aws
Search
[Alt+S]
Europe (Stockholm)
Aarti Gunjal (8948-2163-8224)
Aarti Gunjal
EC2 > Instances > i-0d5e01d4e38ed6ae4 > Create image
Create image Info
An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.
Image details
Instance ID
i-0d5e01d4e38ed6ae4 (websystem)
Image name
systemImage
Maximum 127 characters. Can't be modified after creation.
Image description - optional
Image description
Maximum 255 characters
Reboot instance


Gmail
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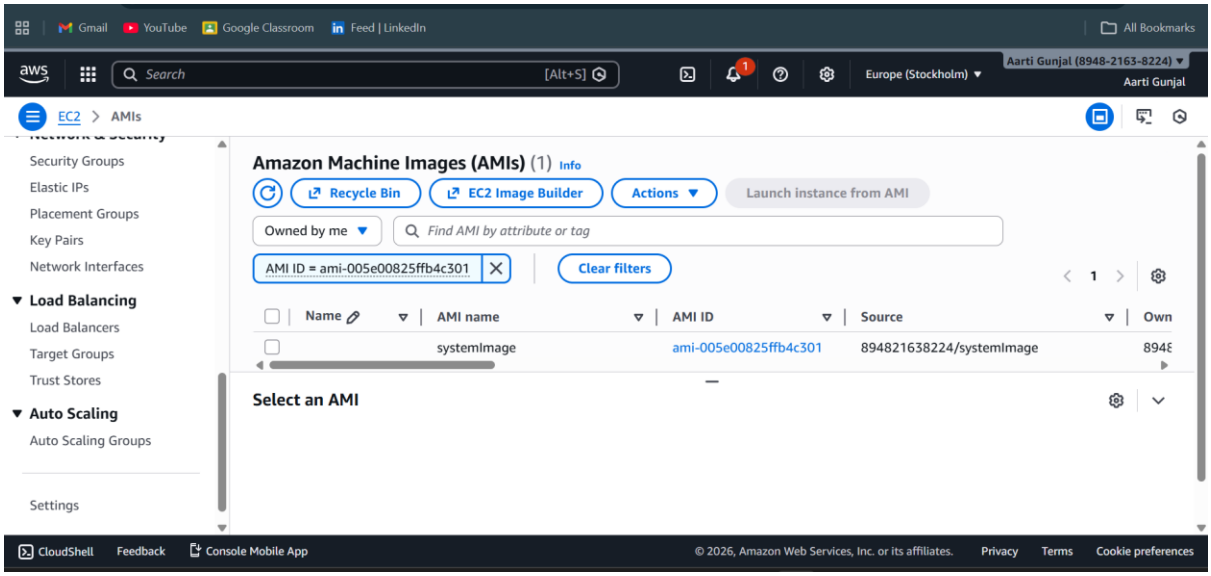
Europe (Stockholm)
Aarti Gunjal (8948-2163-8224)
Aarti Gunjal
EC2 > Instances > i-0d5e01d4e38ed6ae4 > Create image
Add volume
During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Tag image and snapshots together
Tag the image and the snapshots with the same tag.
Tag image and snapshots separately
Tag the image and the snapshots with different tags.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Create image
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

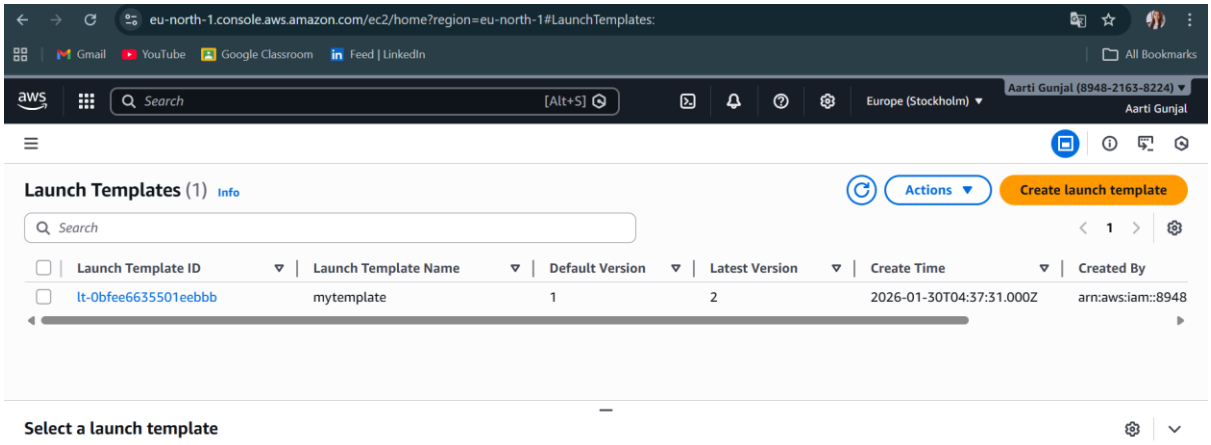


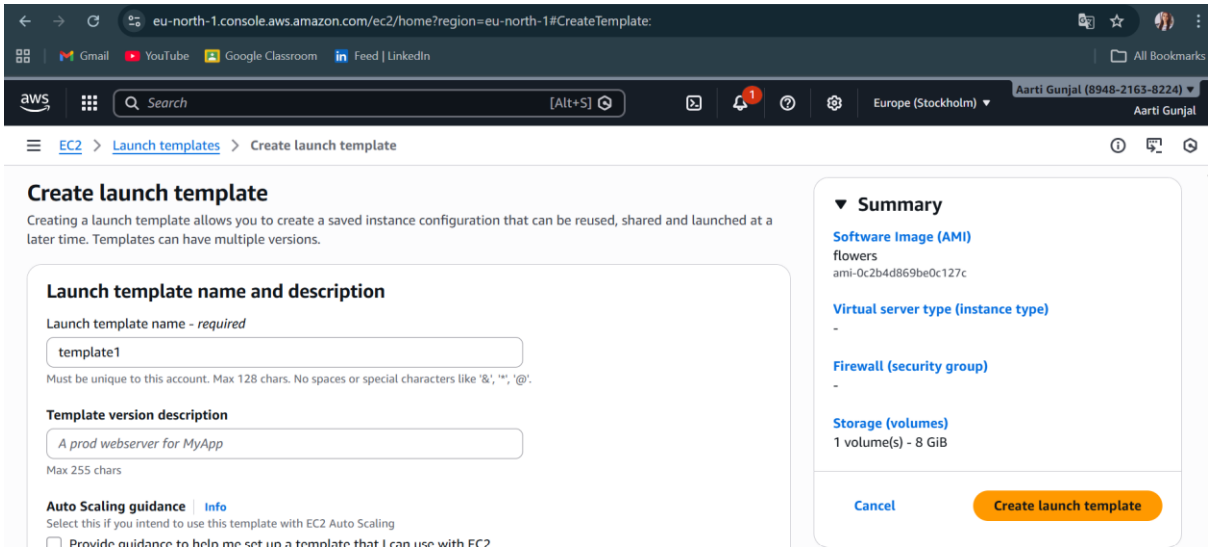
3. Create a Launch Template

Launch Template defines **how new EC2 instances will be created**.

Steps:

1. Go to **EC2 → Launch Templates**
2. Click **Create Launch Template**
3. Provide:
   - Launch Template Name
   - Version description
4. Select:
   - AMI (created in Step 2)
   - Instance type
   - Key pair
   - Security group
5. Click **Create Launch Template**





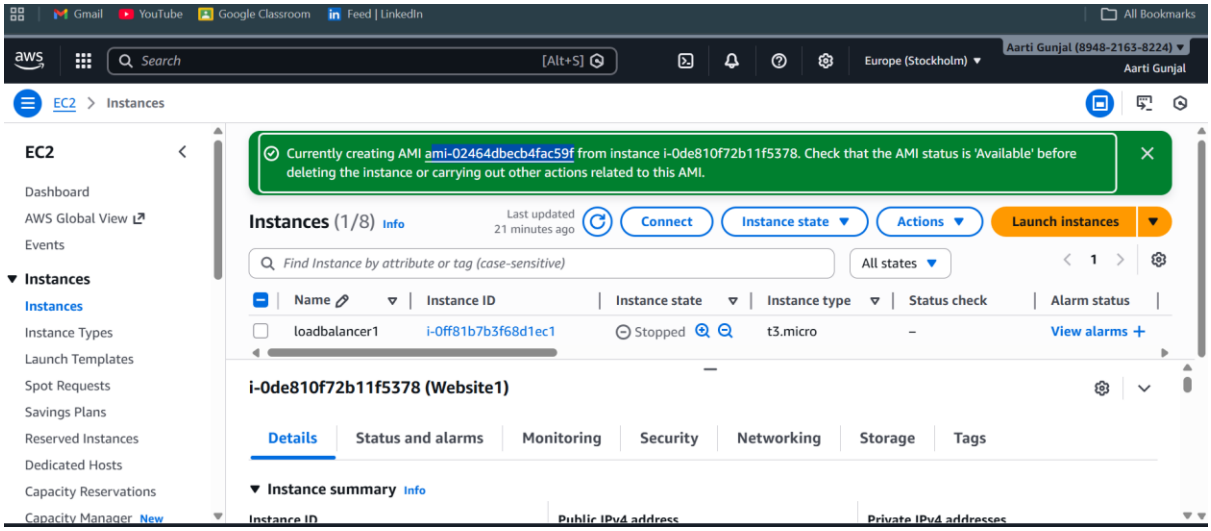
4. Create an Application Load Balancer (ALB)

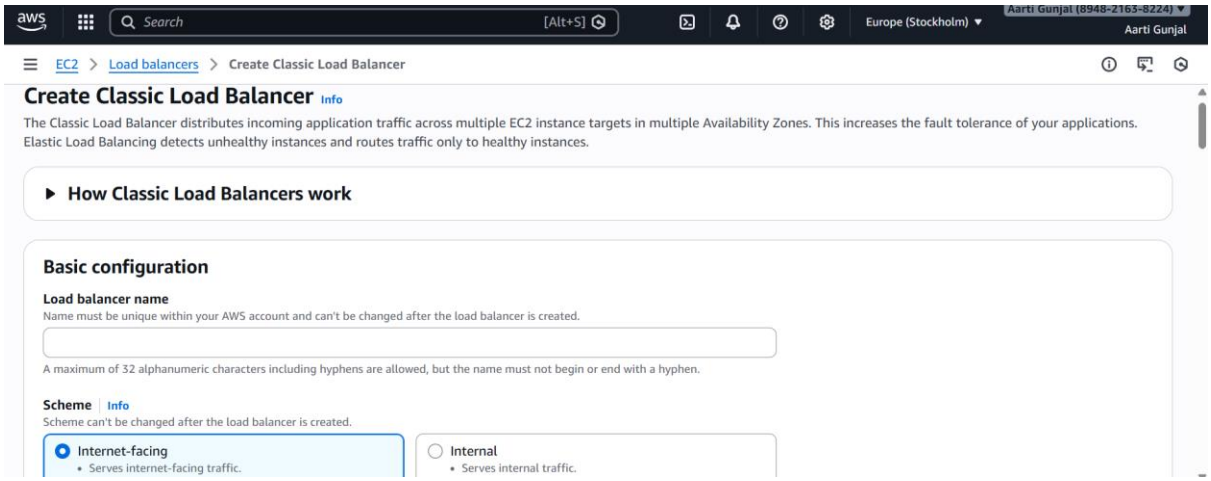
Load Balancer distributes traffic across multiple EC2 instances.

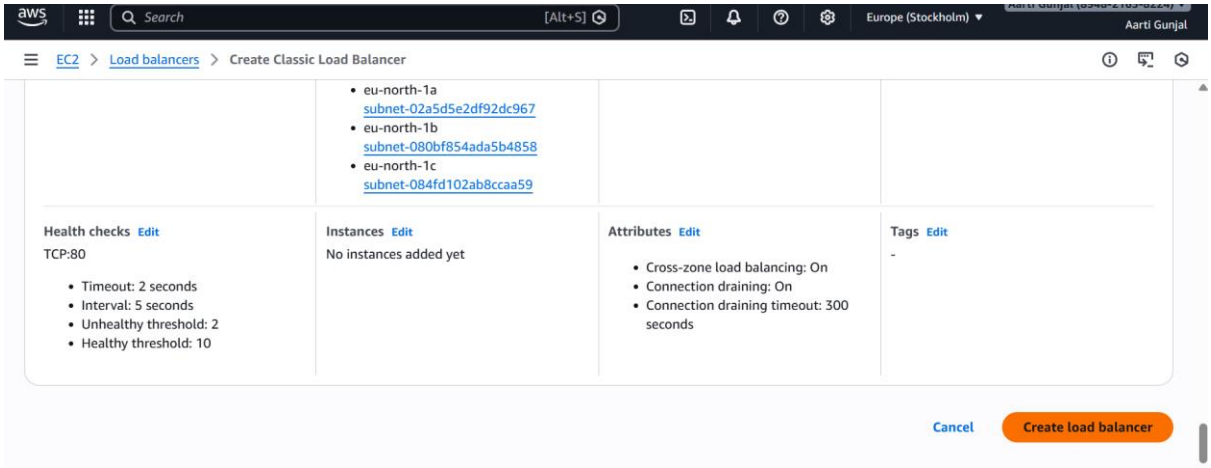
Steps:

1. Go to **EC2 → Load Balancers**
2. Click **Create Load Balancer**
3. Select **Application Load Balancer**
4. Configure:
   - Name
   - Scheme: Internet-facing
   - IP type: IPv4
5. Select:
   - VPC
   - At least 2 Availability Zones

6. Create **Target Group**:
    - Target type: Instance
    - Protocol: HTTP
    - Port: 80
7. Register instances
8. Click **Create Load Balancer**





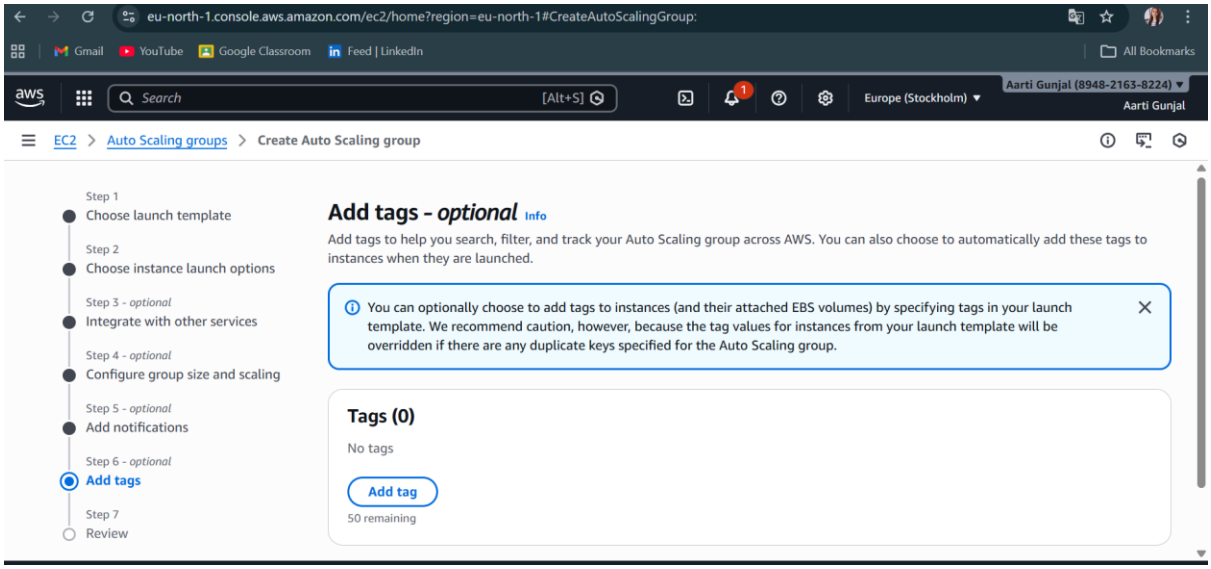


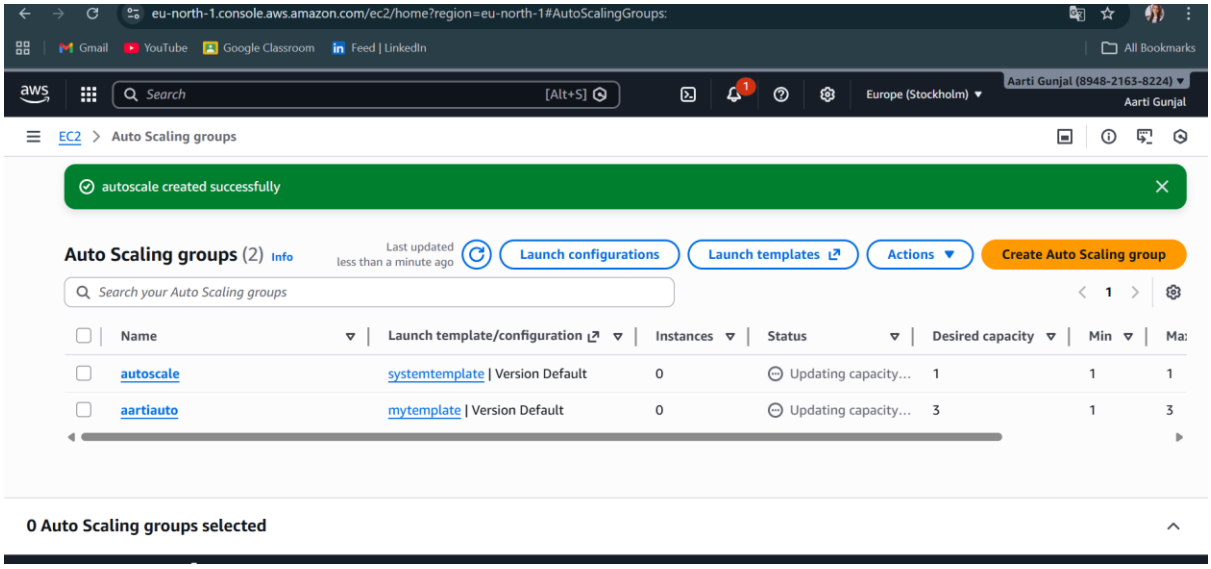
5. Create Auto Scaling Group (ASG)

Auto Scaling Group automatically increases or decreases EC2 instances based on load.

Steps:

1. Go to **EC2 → Auto Scaling Groups**
2. Click **Create Auto Scaling Group**
3. Enter:
   - Auto Scaling Group Name
4. Select:
   - Launch Template (created in Step 3)
5. Choose:
   - VPC
   - Subnets (minimum 2 AZs)
6. Attach Load Balancer:
   - Select existing ALB
   - Choose Target Group
7. Configure Group Size:
   - Desired capacity: 2
   - Minimum capacity: 1
   - Maximum capacity: 4




eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#AutoScalingGroups:
Gmail
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Europe (Stockholm)
Aarti Gunjal (8948-2163-8224)
Aarti Gunjal
EC2 > Auto Scaling groups
autoscale created successfully
Auto Scaling groups (2) Info
Last updated less than a minute ago
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Search your Auto Scaling groups
Name
Launch template/configuration
Instances
Status
Desired capacity
Min
autoscale
systemtemplate | Version Default
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Updating capacity...
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Updating capacity...
3
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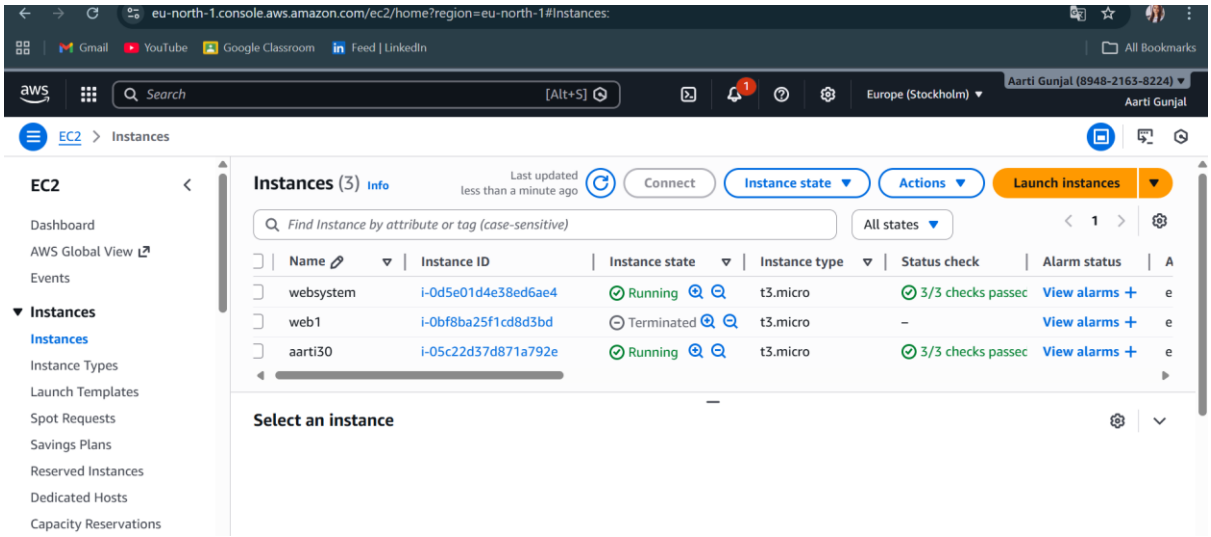
3
0 Auto Scaling groups selected


eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:
Gmail
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Google Classroom
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All Bookmarks
Search
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Europe (Stockholm)
Aarti Gunjal (8948-2163-8224)
Aarti Gunjal
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Instances (3) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
websystem
i-0d5e01d4e38ed6ae4
Running
t3.micro
3/3 checks passed
View alarms
web1
i-0bf8ba25f1cd8d3bd
Terminated
t3.micro
View alarms
aarti30
i-05c22d37d871a792e
Running
t3.micro
3/3 checks passed
View alarms
Select an instance